Muhammad Al Thariq Fairuz

Machine Learning Engineer

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Informatics Engineering graduate (cum laude, GPA 3.6/4.0) from Institut Teknologi Bandung with hands-on experience in machine learning research and in deploying AI systems in practice. Presented research at IEEE ICAICTA 2026 and worked on ML for real-world engineering problems in a cross-disciplinary team. Recognised in national and international competitions for applied AI. Experienced across the full ML lifecycle, from data preparation and modelling to building and shipping working applications. Looking to contribute to research or industry teams where machine learning is used to solve meaningful, data-driven problems.

Work Experiences

Juxtalabs **April 2026 - Present**

AI Engineer

Deployed as part of a 3-person engineering team to optimize Bank Mega's telemarketing QC platform, building an automated system to assess and score sales agents' performance. Reduced data processing latency by up to 98% and streamlined the QA workflow to handle high daily volumes of sales calls with near-zero manual intervention

Technologies and expertise: AI Engineering, Software Engineering, Python, OpenAI, PostgreSQL, Redis, MinIO, RAG.

Institut Teknologi Bandung **April 2026 - Present**

Research Assistant

Developed computer vision models to detect and classify road damage in satellite and aerial imagery for post-disaster road network restoration planning, in collaboration with civil engineering researchers at ITB and ITERA.

Technologies and expertise: Machine Learning, Huggingface, Python, QGIS, PyTorch

PT Bati Investasi Teknologi **October 2025 - January 2026**

AI Software Engineer

Collaborated with the engineering team to develop and maintain scalable software solutions, ensuring high code quality and performance. Architecting and developing a RAG Chatbot to enhance data retrieval capabilities.

Technologies and expertise: Web Development, Software Engineering, FastAPI, Typescript, PostgreSQL, Qdrant

PT Kilang Pertamina Internasional **July 2025 - September 2025**

AI Engineer

Developed proof-of-concept models for equipment predictive maintenance using limited data, including an autoencoder-based anomaly detector (F1-score > 0.75) and a forecasting model built on Amazon Chronos (WAPE < 8%), and delivered the results through an ASP.NET interface for end users.

Technologies and expertise: Data Science, Deep Learning, Software Engineering, Tensorflow, C#, ASP.NET, MongoDB

Institut Teknologi Bandung, Bandung, Indonesia **September 2024 - June 2026**

Artificial Intelligence Lab Assistant

Provided comprehensive mentorship and collaborated with professors to redesign practical assignments to undergraduate students across three computer science courses: Computational Logic, Artificial Intelligence, and Machine Learning.

Technologies and expertise: Python, Jupyter Notebook, Matplotlib, Seaborn, Scikit-learn, PyTorch, Tensorflow, Deep Learning

Achievements

Finalist **October 2026**

HACKaSTONE 2026

Built an agentic AI learning platform for deaf and hard-of-hearing learners, featuring real-time sign-language scoring, a two-way sign/speech translation bridge with a 3D signing avatar, and AI-generated accessible lessons and quizzes. Selected as a Finalist for the final round in Amsterdam.

1st Place Hackathon **June 2026**

Gunadarma Code Week 2026

Developed Tukangin, an AI-powered home service platform that bridges customers with trusted skilled workers for home repair and maintenance.

Innovation Award — May 2026

BRICS Industrial Innovation Contest 2026

Received the Innovation Award for developing SEAGuard, AI-Driven Intelligence for Regional Safety.

1st Honorable Mention Hackathon — January 2026

Dinus Application Competition 2026

Achieved an Honorable Mention (4th place) at the DINACOM 2026 Hackathon, a 12-hour competition, where we built SahabatGuru, a teacher assistant platform that automates exam creation, helps design learning materials and lesson plans (RPP), proctors exams, and analyzes student performance.

Publications & Intellectual Property

RestoraX — October 2026

M. A. Fairuz, R. Verdian, D. Dharma, F. Zukhruf, et al. Copyright holder: Institut Teknologi Bandung. Reg. No. 001514719, Ministry of Law, Republic of Indonesia, 2026.

Developed computer vision models to detect and classify road damage from satellite and aerial imagery for RestoraX, supporting post-disaster road network restoration planning in collaboration with civil engineering researchers at ITB and ITERA. The software is registered as a computer-program copyright in Indonesia (Reg. No. 001514719).

Multi-Agent Reinforcement Learning for Portfolio Management on the Indonesia Stock Exchange: A Microstructure-Aware Ablation Study — October 2026

Muhammad Al Thariq Fairuz, Dody Dharma. IEEE ICAICTA 2026, Malaysia (presented at IEEE ICAICTA 2026)

Built a multi-agent reinforcement learning system for 53 Indonesian stocks on 5-minute data, blending five SAC agents through a PPO meta-controller in a simulator of IDX trading rules (ARA/ARB limits, asymmetric costs, lot sizes). Returned +7.23% over eleven walk-forward windows versus LQ45 at −12.89%, and limited losses to −19.11% on a bear-market holdout (LQ45 −23.22%). Ablations showed the gains come from policy diversification rather than learned regime adaptation.

Education

Institut Teknologi Bandung, Indonesia — 2022 - 2026

B.Eng. in Informatics Engineering, Cum Laude, GPA 3.6/4.0

Thesis: Multi-Agent Reinforcement Learning for Portfolio Management on the Indonesia Stock Exchange with Market Microstructure Modeling (Advisor: Dody Dharma)